

HARSH RAJ THAKUR

Researcher at IIT Roorkee

@ kspqa126@gmail.com

+91 7550233792

IIT Roorkee, Uttarakhand, 247667, India

Roorkee, India

https://www.linkedin.com/in/harsh-raj-thakur-a52790114/



PUBLICATIONS

- [1] Harsh Raj Thakur, Sparsh Mittal, A Transistor-Level Implementation of Radix-2 and Radix-4 FFT Using Logical Effort, IEEE VLSID 2026, Pune, India.
- [2] Harsh Raj Thakur, Sparsh Mittal, Design and ASIC Implementation of a Novel Three-Stage Pipelined Posit(16,2) Multiplier, Submitted to IEEE ISVLSI 2026.
- [3] Harsh Raj Thakur, Seenivasn M A, Prabir Saha, VLSI Implementation of Convolution Chip: A Design Prospects, International Conference Microsystem Technology 2023(Micro-2023).
- [4] Rekib Uddin Ahmed, Harsh Raj Thakur, Seenivasn M A, Prabir Saha, Power-Efficient VLSI Realization of Decimal Convolution Algorithms for Resource-Constrained Environments: A Design Perspective in CMOS and Double-Gate CMOS Technology, Microsystem Technology, Springer, 2024, (Journal)Published.
- [5] Harsh Raj Thakur, Phannong Youngkuk, A Comparative Analysis of Power Consumption and Delay in 4-bit Vedic Multiplier and a Proposed Multiplier Utilizing Variable Bit Adders for CNN Logic, Accepted in IEEE International Conference-INDISCON 2024.
- [6] Harsh Raj Thakur, Prabir Saha, VLSI Implementation of Edge Detection Chip: A Prospective Design, accepted and presented in ISES - 2024.

PROJECTS

Two-stage operational amplifier Simulation and Layout on Cadence

Research 1st Semester Project/IIT Roorkee

- Designed and characterized a two-stage operational amplifier with Miller compensation in Cadence SPECTRE. Achieved differential gain >70 dB, GBP >100 MHz, phase margin >60°, and slew rate ≥ 0.1 V/ns. Optimized the design for high CMRR, low power dissipation, and operation within a 1.8 V supply voltage. Validated performance through AC, transient, and RC-extracted layout simulations.

Transistor-Level FFT Design using Logical Effort

Research 3rd Semester Project/IIT Roorkee

- Designed Radix-2 and Radix-4 FFT architectures at transistor level using logical effort methodology.
- Optimized critical path delay and power using custom 28T full adder design.
- Improved throughput and speed for real-time DSP applications through efficient gate sizing and pipeline optimization.

Three-Stage Pipelined Posit(16,2) Multiplier ASIC Design

EDUCATION

M.Tech VLSI & EMBEDDED SYSTEM
NIT MEGHALAYA (GPA 9.2)

August 2022 – May 2024

B.E ELECTRICAL AND ELECTRONICS ENGINEERING

SATHYABAMA UNIVERSITY

AUGUST 2014 – June 2018

EXPERIENCE

Research Scholar

IIT Roorkee (GPA 8)

13 July 2024

Current Position

Intern

Maven Silicon

30 Mar 2022 - 05-Oct-2022

- RTL (Register Transfer Level) Designer
- Cadence Simulation, Layout, Innovus, genus
- Report Writing

Placement Coordinator

NIT Meghalaya

25 Dec 2022 – 25 Aug 2024

- Management
- Internship, Placement Drives.
- Collaboration

REFEREES

Associate Prof. Dr. Sparsh Mittal Kumar

@ https://www.iitr.ac.in/

sparsh.mittal@ece.iitr.ac.in

Indian Institute of Technology, Roorkee

Associate Prof. Dr. Prabir Kumar Saha

@ https://www.nitm.ac.in/

prabir.saha@nitm.ac.in

Research 4th semester Project/IIT Roorkee

- Designed and implemented a pipelined Posit(16,2) multiplier for high-performance arithmetic.
- Optimized latency and throughput using three-stage pipeline architecture.
- Targeted DSP and AI workloads with improved accuracy and hardware efficiency over IEEE floating-point.

DesignTechVLSI – Technical Platform for VLSI Learning
Personal Project

- Developed and deployed **DesignTechVLSI** (designtechvlsi.com), a platform for VLSI and semiconductor learning.
- Built full-stack architecture using Firebase and Render for back-end services, hosting, and real-time data handling.
- Developed REST APIs for dynamic content delivery and seamless user interaction.
- Implemented Firebase for authentication, database management, and analytics monitoring.
- Deployed backend on Render with continuous deployment and performance monitoring.

Power-Efficient VLSI Realization of Decimal Convolution Algorithms towards Edge Detection

M.Tech Thesis Project/NIT Meghalaya

- Focused on designing power-efficient VLSI circuits for the implementation of decimal convolution algorithms used in edge detection. The project aimed to optimize power consumption while maintaining high performance, targeting applications in image processing and computer vision. Contributions included circuit design, optimization techniques, and simulation to achieve efficient and reliable edge detection functionality.

Neuromorphic Synapse Circuit Design and MOSFET Sizing with Cadence Tools

M.Tech 2nd Semester Project/NIT Meghalaya

- Designed and simulated synapse circuits for the "Neuromorphic Computing Architectures, Models, and Applications" project using Cadence tools. Developed a spike generator with PWM and op-amp techniques and optimized MOSFET sizing using the gm/Id methodology to advance neuromorphic computing for AI applications.

Comparative Analysis of Multipliers in FPGA using Verilog on Xilinx

M.Tech 1st Semester Project/NIT Meghalaya

- Analyzed the performance of different 8-bit multipliers (conventional, Vedic, and hybrid) implemented in Verilog on Xilinx FPGA. Compared efficiency, speed, and resource utilization to evaluate practical applications.

TOOLS/KEY SKILLS

Verilog

xilinx

Model sim

cadence

Virtusoo

innovus

genus

Windows

Centos

linux

MS-Excel

LANGUAGES

English

Hindi

Japanese